

Mammals from Tamaulipas, Mexico

Rollin H. Baker

The background of the lower half of the cover is a vibrant blue field filled with a complex, abstract pattern of yellow geometric shapes. These shapes include various sizes of triangles, diamonds, squares, and curved segments, some of which are nested or overlapping, creating a dynamic and modern visual texture.

Project Gutenberg

Mammals from Tamaulipas, Mexico

Rollin H. Baker

A decorative pattern consisting of various geometric shapes in magenta and blue, including triangles, diamonds, and lines, arranged in a complex, overlapping manner.

Project Gutenberg

The Project Gutenberg eBook of Mammals from Tamaulipas, Mexico

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: Mammals from Tamaulipas, Mexico

Author: Rollin H. Baker

Release date: February 28, 2010 [eBook #31448]

Most recently updated: January 6, 2021

Language: English

Other information and formats: www.gutenberg.org/ebooks/31448

Credits: Produced by Chris Curnow, Joseph Cooper, Diane Monico, and the Online Distributed Proofreading Team at <https://www.pgdp.net>

***** START OF THE PROJECT GUTENBERG EBOOK MAMMALS
FROM TAMAULIPAS, MEXICO *****

Mammals from Tamaulipas, Mexico

BY

ROLLIN H. BAKER

**University of Kansas Publications
Museum of Natural History**

**Volume 5, No. 12, pp. 207-218
December 15, 1951**

**University of Kansas
LAWRENCE
1951**

UNIVERSITY OF KANSAS PUBLICATIONS, MUSEUM OF NATURAL HISTORY

Editors: E. Raymond Hall, Chairman, A. Byron Leonard,
Edward H. Taylor, Robert W. Wilson

Volume 5, No. 12, pp. 207-218
December 15, 1951

UNIVERSITY OF KANSAS
Lawrence, Kansas

PRINTED BY
FERD VOILAND, JR., STATE PRINTER
TOPEKA, KANSAS
1951

23-8338

Mammals from Tamaulipas, Mexico

By

ROLLIN H. BAKER

Forming the northeastern border of Mexico, Tamaulipas extends in an elongated, north-south direction from the Temperate into the Torrid Zone and contains faunal elements from both the Nearctic and Neotropical regions. The mammals are less known than those from some of the bordering states; for the most part collecting has been limited to a few localities, chiefly along the Pan-American Highway. Accordingly, as a step towards a long-range study of the mammals of Tamaulipas, the Museum of Natural History of the University of Kansas acquired from William J. Schaldach, Jr., a small, but significant, collection of mammals taken in the last month of 1949 and the first four months of 1950.

Collections were made at several places in the vicinity of Ciudad Victoria, including localities along the humid, eastern face of the Sierra Madre Oriental. Many of these specimens were obtained near camps made west of the village of El Carrizo. This small community is on the Pan-American highway, 70 kilometers (by highway) south of Ciudad Victoria. The resulting collections, which are reported upon here, disclose that several tropical mammals range farther northward than previously reported. Funds for financing the field work were made available by a grant from the Kansas University Endowment Association.

ACCOUNTS OF SPECIES

***Didelphis mesamericana mesamericana* Oken**

Central American Opossum

Did[elphys] mes-americana Oken, Lehrbuch d. Naturgesch., pt. 3, vol. 2:1152, 1816. (Type from Northern Mexico.)

Didelphis mes-americana Allen, Bull. Amer. Mus. Nat. Hist., 16:256, August 18, 1902.

Specimens examined, 2 as follows: 36 km. N and 10 km. W Ciudad Victoria, 1 km. E El Barretal, on Río Purificación, 1; 12 km. N and 4 km. W Ciudad Victoria, 1.

***Philander opossum pallidus* (Allen)**

Four-eyed Opossum

Metachirus fuscogriseus pallidus Allen, Bull. Amer. Mus. Nat. Hist., 14:215, July 3, 1901. (Type from Orizaba, Veracruz, Mexico.)

Philander opossum pallidus Dalquest, Occ. Papers Mus. Zool., Louisiana State Univ., No. 23:2, July 10, 1950.

Specimens examined, 3 from 70 km. [by highway] S Ciudad Victoria and 2 km. W El Carrizo.

Remarks.—These three specimens have proportionately longer tails than typical *P. o. pallidus* from central Veracruz; total length and length of tail of two adult males are 575, 295, and 568, 290 respectively.

This marsupial has been previously unrecorded from Tamaulipas or from so northward a locality. The four-eyed opossum evidently ranges northward along the east face of the Sierra Madre Oriental within the humid division of the Upper Tropical Life-zone. These animals, all males, were taken in steel traps baited with the bodies of skinned mice or birds. Sets were made along well-used trails leading from a densely vegetated arroyo into a corn field through openings in a fence of roughly piled logs. The elevation of this locality is approximately 2500 feet.

***Desmodus rotundus murinus* Wagner**

Vampire Bat

D[esmodus] murinus Wagner, Schreber's Säugthiere, Suppl., 1:377, 1840. (Type from Mexico.)

Desmodus rotundus murinus Osgood, Field Mus. Nat. Hist., publ. 155, zool. ser., 10:63, January 10, 1912.

Specimens examined, 9 as follows: 12 km. W and 8 km. N Ciudad Victoria, 2500 ft., 3; 70 km. [by highway] S Ciudad Victoria and 6 km. W of the [Pan-American] highway [at El Carrizo], 6.

Remarks.—Vampire bats were taken at two caves. At the cave called "Los Troncones", 12 kilometers west and 8 kilometers north of Ciudad Victoria, seven bats were shot down; three were saved. The second cave, south of Ciudad Victoria, was considerably damper than the first. Vampires were found in a small side chamber; nine bats were knocked down. No other kinds of bats were present in either cave.

***Sylvilagus floridanus connectens* (Nelson)**

Florida Cottontail

Lepus floridanus connectens Nelson, Proc. Biol. Soc. Washington, 17:105, May 18, 1904. (Type from Chichicastle, Veracruz, Mexico.)

Sylvilagus floridanus connectens Lyon and Osgood, Catal. Type spec. Mamm. U. S. Nat. Mus., Bull. 62:32, January 28, 1909.

Specimens examined, 2 from 70 km. [by highway] S. Ciudad Victoria and 2 km. W El Carrizo.

***Sciurus aureogaster aureogaster* F. Cuvier**

Red-bellied Squirrel

[*Sciurus*, by implication] *aureogaster* F. Cuvier, Hist. nat. mammifères, vol. 6, livr. 59, pl. with text, September, 1829.

Binomial published at end of work only, vol. 7, tabl. generale et methodique, p. 4, 1842. (Type from "California" = eastern Mexico.)

Specimens examined, 13 as follows: 43 km. S Ciudad Victoria, 1; 70 km. [by highway] S Ciudad Victoria, and 6 km. W of [Pan-American] highway [at El Carrizo], 3; 70 km. [by hwy.] S Victoria and 2 km. W El Carrizo, 9.

Remarks.—These colorful squirrels were taken in thick timber and brush, and locally are referred to as "ardilla pinta" or "ardilla colorada." One female obtained on January 22 is black. Local hunters state that these squirrels are most active early in the morning and late in the afternoon. Two squirrels were seen in copulation on January 19.

***Sciurus negligens* Nelson**

Little Gray Squirrel

Sciurus negligens Nelson, Proc. Biol. Soc. Washington, 12:147, June 3, 1898. (Type from Alta Mira, Tamaulipas, Mexico.)

Specimens examined, 17 as follows: 70 km. [by highway] S of Ciudad Victoria and 6 km. W of the [Pan-American] highway [at El Carrizo], 5; 70 km. [by highway] S Ciudad Victoria and 2 km. W El Carrizo, 12.

Remarks.—Referred to as "ardilla chica" locally, these squirrels were most frequently seen in the dense forest of the lower elevations. Active both in trees and on the ground, these animals were reported as being destructive to corn crops. Females taken in January were lactating.

***Heterogeomys hispidus concavus* Nelson and Goldman**

Hispid Pocket Gopher

Heterogeomys hispidus concavus Nelson and Goldman, Proc. Biol. Soc. Washington, 42:148, March 30, 1929. (Type from

Pinal de Amoles, Queretaro, Mexico.)

Specimens examined, 5 as follows: 70 km. [by highway] S Ciudad Victoria and 5 km. W El Carrizo, 4; 70 km. [by highway] S Ciudad Victoria and 2 km. W El Carrizo, 1.

Remarks.—These gophers have been compared with specimens of *H. h. hispidus* from Veracruz (5 km. N Jalapa and 4 km. WNW Fortin) and with specimens of *H. h. concavus* from San Luis Potosí (Xilitla and vicinity and 3 mi. NW Pujal); the latter were examined through the courtesy of Dr. George H. Lowery, Jr., of the Museum of Zoology at Louisiana State University. These five specimens are assigned to *H. h. concavus* and resemble in every way this subspecies except: slightly smaller, somewhat darker, and skull with lambdoidal crest less inclined forward. In the latter feature, the single skull available seems to resemble most closely that of *H. h. hispidus*. Unfortunately, all but one skull, that of a subadult female, were destroyed in the field.

This is the first known record of this genus in Tamaulipas and the most northern locality from which specimens have been taken. At this latitude, the gopher appears to be restricted to the humid belt on the east face of the Sierra Madre Oriental. These large gophers were difficult to trap by ordinary means; Schaldach got two by using large-sized Macabee traps but the others were taken at night either with the aid of a dog or by natives with slingshots.

***Liomys irroratus texensis* Merriam**

Spiny Pocket Mouse

Liomys texensis Merriam, Proc. Biol. Soc. Washington, 15:44, March 5, 1902. (Type from Brownsville, Cameron County, Texas.)

Liomys irroratus texensis Goldman, N. Amer. Fauna, 34:59, September 7, 1911.

Specimens examined, 15 as follows: 7 km. S and 2 km. W San Fernando, 8; 36 km. N and 10 km. W Ciudad Victoria, 1 km. E El Barretal, on Río Purificación, 1; 70 km. [by highway] S Ciudad Victoria and 2 km. W El Carrizo, 6.

Remarks.—Most of these mice were taken in densely vegetated fallow fields, where both grass and brush were found. Many of the mice were captured at their burrow openings, some of which were found to be plugged and others not plugged.

***Reithrodontomys fulvescens intermedius* Allen**

Fulvous Harvest Mouse

Reithrodontomys mexicanus intermedius Allen, Bull. Amer. Mus. Nat. Hist., 7:136, May 21, 1895. (Type from Brownsville, Cameron County, Texas.)

Reithrodontomys fulvescens intermedius Howell, N. Amer. Fauna, 36:47, June 5, 1914.

Specimen examined, 1 from 70 km. [by highway] S Ciudad Victoria and 2 km. W El Carrizo.

***Baiomys taylori taylori* (Thomas)**

Pygmy Mouse

Hesperomys (Vesperimus) taylori Thomas, Ann. and Mag. Nat. Hist, ser. 5, 19:66, January, 1887. (Type from San Diego, Duval County, Texas.)

Baiomys taylori Mearns, Mamm. Mex. Bound., Bull. 56:381, April 13, 1907.

Specimens examined, 2 from 70 km. [by highway] S Ciudad Victoria and 2 km. W El Carrizo.

Remarks.—Specimens were captured in runways in dense grass and weeds at the edge of a corn field.

***Peromyscus leucopus texanus* (Woodhouse)**

White-footed Mouse

Hesperomys texana Woodhouse, Proc. Acad. Nat. Sci. Philadelphia, 6:242, 1853. (Type probably from the vicinity of Mason, Mason County, Texas.)

Peromyscus leucopus texanus Osgood, N. Amer. Fauna, 28:127, April 17, 1909.

Specimens examined, 8 as follows: 7 km. S and 2 km. W San Fernando, 1; 12 km. N and 4 km. W Ciudad Victoria, 2; 70 km. [by highway] S Ciudad Victoria and 2 km. W El Carrizo, 4.

***Peromyscus ochraventer* new species**

Brown-bellied Wood Mouse

Type.—Female, adult, skin and skull; no. 36958, Univ. Kansas Mus. Nat. Hist.; 70 km. [by highway] S Ciudad Victoria and 6 km. W of the [Pan-American] highway [at El Carrizo], Tamaulipas, Mexico; 12 January 1950; obtained by William J. Schaldach, Jr., original no. 566.

Range.—Known only from the type locality; probably found in other localities along the humid, east face of the Sierra Madre Oriental in Tamaulipas.

Diagnosis.—Size medium (see measurements); upper parts near Ochraceous Tawny (capitalized color terms after Ridgway, Color Standards and Color Nomenclature, Washington, D. C., 1912), brighter on sides and duller on back; cheeks, sides of neck, shoulders and upper forelegs lighter, between Ochraceous Buff and Ochraceous Orange; eye ring dark; underparts light

Cinnamon Buff, breast patch brighter; ears dusky, sparsely covered with hairs colored like back; feet white; tail scaly in appearance, indistinctly bicolored with short dark hairs above and short pale hairs below; skull without beaded or ridged supraorbital border; rostrum expanded anteriorly with sides almost parallel; teeth with strongly developed outer accessory cusps on the first and second upper molar teeth; anteriormost loph (parastyle-protoconule of Goldman, N. Amer. Fauna, 43:11, September 23, 1918) of the first upper molar large, almost as broad as greatest breadth of tooth.

Comparisons.—*Peromyscus ochraventer* has been compared with *P. difficilis* (specimens from Veracruz), *P. boylei* (Veracruz), *P. banderanus* (Guerrero), *P. mexicanus* (Veracruz), *P. furvus* (Veracruz), and *P. latirostris* (San Luis Potosí). From *P. difficilis*, *P. ochraventer* differs in having underparts distinctively brownish, rostrum expanded anteriorly with sides almost parallel, anteriormost loph of the first upper molar larger, and auditory bulla smaller. From *P. boylei*, *P. ochraventer* differs in having underparts distinctively brownish, tail less distinctly bicolored, rostrum expanded anteriorly with sides almost parallel, and anteriormost loph of the first upper molar larger. From *P. banderanus*, *P. ochraventer* differs in having underparts distinctively brownish, tail less distinctly bicolored, rostrum expanded anteriorly with sides almost parallel, anteriormost loph of the first upper molar larger, auditory bulla smaller, and in lacking a beaded or ridged supraorbital border. From *P. mexicanus*, *P. ochraventer* differs in having underparts distinctively brownish, tail not irregularly blotched with dusky, rostrum expanded anteriorly with sides almost parallel, anteriormost loph of the first upper molar larger, and in lacking a beaded or ridged supraorbital border. From *P. furvus* and *P. latirostris*, *P. ochraventer* differs in being smaller, having underparts distinctively brownish, tail not irregularly blotched with dusky, rostrum proportionately shorter, and interpterygoid space relatively narrower.

Remarks.—*Peromyscus ochraverter* is considered to be a distinct species showing little evident relationship with other Mexican *Peromyscus*. In the shape of the skull, especially the anterior expansion of the rostrum, *P. ochraverter* seems to be related to *P. furvus* and *P. latirostris*, a series of the latter being made available for examination by Dr. George G. Lowery, Jr., of the Museum of Zoology at Louisiana State University. However, the rostrum of these two larger species is proportionately longer than the rostrum of *P. ochraverter*. In size, coloration and most cranial features, *P. ochraverter* resembles *P. mexicanus*, although the absence, instead of presence, of a supraorbital bead or ridge, the almost parallel-sided, instead of more pointed, rostrum and the larger, instead of smaller, anteriormost loph of the first upper molar in *P. ochraverter* are well-marked differences. The baculum of *P. ochraverter* is much shorter with a proportionately heavier base and shaft than the baculum of *P. mexicanus mexicanus* (from Veracruz) and *P. m. saxatilis* (from Costa Rica). The geographic range of *Peromyscus ochraverter* is not known to meet that of *P. mexicanus*; the nearest place to the type locality of *P. ochraverter* from which *P. mexicanus* has been taken is at Xilitla approximately 225 kilometers to the southward in San Luis Potosí (Dalquest, Occ. Papers Mus. Zool., Louisiana State Univ., No. 28:8, July 10, 1950).

The brown coloring on the underparts is a distinctive feature of *P. ochraverter*; in adults this color differs in shade. In some specimens patches of whitish hair give the tail a splotched appearance. Eleven of the twenty-eight skulls and lower jaws examined have bone eroded away from around the cheek-teeth exposing part of the roots. Most of the fully adult animals have this condition. One adult female, no. 36959, has the upper third molar on the right side missing, possibly as a result of bone erosion.

These mice were taken in junglelike forest, in rocks and adjacent to logs. Schaldach writes that "Their burrows go back under the large limestone blocks, and each burrow where I caught a mouse has a pile of excavated earth, like a tiny gopher mound." The trapping area was at an elevation of approximately 2800 feet on the steep sides of a small hill on top of which the field camp was situated. Schaldach indicated that this locality was transitional between arid tropical and humid tropical conditions. On January 13, 1950, a female taken was lactating and had five recent placental scars; another taken the same day also had five placental scars.

Measurements.—Average and extreme measurements of seven adult males and six adult females of *P. ochraverter* from the type locality are, respectively, as follows: Total length, 238.0 (227-249), 236.5 (226-248); length of tail, 124.4 (117-127), 122.2 (116-128); length of hind foot, 25.6 (24-26), 25.5 (25-26); length of ear from notch, 20.9 (20-21), 20.7 (20-21); greatest length of skull, 31.0 (30.6-31.9), 30.8 (30.5-31.0); basilar length, 23.3 (22.7-23.8), 23.4 (23.0-23.9); zygomatic breadth, 15.1 (14.6-15.7), 15.0 (14.9-15.2); post palatal length, 10.6 (10.5-10.9), 10.9 (10.5-11.2); interorbital breadth, 4.7 (4.5-4.9), 4.7 (4.6-4.8); mastoidal breadth, 12.8 (12.4-13.2), 12.8 (12.6-12.9); length of nasals, 11.6 (10.8-12.0), 11.6 (11.2-11.7); length of shelf of bony palate, 4.7 (4.5-4.8), 4.6 (4.5-4.7); length of palatine slits, 6.3 (6.0-6.5), 6.2 (6.0-6.4); length of diastema, 8.2 (8.0-8.5), 8.2 (8.1-8.4); alveolar length of upper molariform tooth-row, 4.4 (4.3-4.6), 4.4 (4.3-4.5).

Specimens examined, 28, from the type locality.

***Oryzomys couesi aquaticus* Allen**

Coues Rice Rat

Oryzomys aquaticus Allen, Bull. Amer. Mus. Nat. Hist., 3:289, June 30, 1891. (Type from Brownsville, Cameron County, Texas.)

Oryzomys couesi aquaticus Goldman, N. Amer. Fauna, 43:39, September 23, 1918.

Specimens examined, 2 as follows: 36 km. N and 10 km. W Ciudad Victoria, 1 km. E El Barretal, on Río Purificación, 1; 70 km. [by highway] S Ciudad Victoria and 6 km. W of the [Pan-American] highway [at El Carrizo], 1.

Remarks.—The specimens, all immatures, are slightly darker than topotypes of *O. c. aquaticus*, seemingly tending toward the darker *O. c. peragrus* Merriam to the southward. These records of occurrence extend the

known range of this subspecies approximately 210 miles to the southward and increase the possibility of continuous distribution between *O. c. aquaticus* and *O. c. peragrus*.

The male obtained south of Ciudad Victoria was taken on January 12, by William J. Shaldach, Jr., 200 yards within the tunnel of a mine at an elevation of approximately 2600 feet. This was in the Sierra Gorda, which is a part of the Sierra Madre Oriental.

***Oryzomys rostratus rostratus* Merriam**

Rice Rat

Oryzomys rostratus Merriam, Proc. Washington Acad. Sci., 3:293, July 26, 1901. (Type from Metlaltoyuca, Pueblo, Mexico.)

Specimen examined, 1 from 70 km. [by highway] S Ciudad Victoria and 2 km. W El Carrizo.

Remarks.—This immature male is paler than specimens of *O. r. rostratus* from the state of Veracruz. This locality extends the known range of this species northward a distance of approximately 100 miles. Previously it had been recorded from only as far north as Alta Mira, Tamaulipas (Goldman, N. Amer. Fauna, 43:54, September 23, 1918). This specimen was trapped on February 16 in a rodent runway in dense grass in a fallow cane field.

***Oryzomys fulvescens engraciae* Osgood**

Fulvous Rice Rat

Oryzomys fulvescens engraciae Osgood, Jour. Mamm., 26:300, November 14, 1945. (Type from Hacienda Santa Engracia, northwest of Ciudad Victoria, Tamaulipas, Mexico.)

Specimens examined, 5 from 70 km. [by highway] S Ciudad Victoria and 2 km. E El Carrizo.

Remarks.—These specimens are referred to *O. f. engraciae* on the basis of their pale color and narrow interorbital space. They were taken in runways in dense grass in fallow cane fields.

***Sigmodon hispidus toltecus* (Saussure)**

Hispid Cotton Rat

[*Hesperomys*] *toltecus* Saussure, Revue et magasin de zoologie, ser. 2, 12:98, 1860. (Type from mountains of Veracruz, Mexico.)

Sigmodon hispidus toltecus Bailey, Proc. Biol. Soc. Washington, 15:110, June 2, 1902.

Specimens examined, 23 as follows: 36 km. N and 10 km. W Ciudad Victoria, 1 km. E El Barretal, on Río Purificación, 1; 70 km. [by highway] S Ciudad Victoria and 2 km. W El Carrizo, 22.

Remarks.—Among named kinds of *Sigmodon* this series most closely approaches *S. h. toltecus* to the southward. The specimens are slightly lighter in color of the upper parts than are examples of this same subspecies from 8 km. NW of Potrero, Veracruz, but in other ways are similar. The single specimen from 36 km. N and 10 km. W Ciudad Victoria is a skull only, but seems closest to *S. h. toltecus*. As is often the case with collections of *Sigmodon*, this series contains mostly immatures.

Cotton rats were found abundantly in cultivated areas. Local farmers stated that these rats were destructive to sugar cane by girdling the stems one and one-half inches above the ground.

***Neotoma micropus micropus* Baird**

Baird Wood Rat

Neotoma micropus Baird, Proc. Acad. Nat. Sci. Philadelphia, p. 333, April, 1855. (Type from Charco Escondido, Tamaulipas, Mexico.)

Specimens examined, 2 from 7 km. S and 2 km. W San Fernando.

***Neotoma angustapalata* new species**

Tamaulipan Wood Rat

Type.—Male, subadult, skin and skull; No. 36976, Univ. Kansas Mus. Nat. Hist.; 70 km. [by highway] S Ciudad Victoria and 6 km. W of the [Pan-American] highway [at El Carrizo], Tamaulipas, Mexico; 14 January 1950; obtained by William J. Schaldach, Jr., original no. 578.

Range.—Known only from the type locality; probably found in other localities along the humid, east face of the Sierra Madre Oriental in Tamaulipas.

Diagnosis.—Size large (see measurements); upper parts dusky brown, paler on sides, individual hairs on middle of back tipped with black or with Light Pinkish Cinnamon (capitalized color term after Ridgway, Color Standards and Color Nomenclature, Washington, D. C., 1912); head grayer especially on cheeks; underparts dusky (dark bases of white-tipped hairs exposed), hairs on throat and inguinal region of adult specimen white to base; outside of legs dusky gray; tail scaly in appearance and sparsely covered with short, blackish hairs above and short, whitish hairs below; skull with auditory bulla large; external auditory meatus large; palatine region narrow; sides of interpterygoid fossa concave and broadly excavated near posterior end of molariform tooth-rows.

Comparison.—*Neotoma angustapalata* has been compared with *N. torquata* (specimens from Veracruz and Puebla), *N. navus* (Coahuila), *N. mexicana* (New Mexico), *N. micropus* (Tamaulipas), *N. albigula* (Coahuila), *N. ferruginea* (Jalisco), and *N. distincta* (from published description in Goldman, N. Amer. Fauna, 31:64, October 19, 1910). *Neotoma angustapalata* differs from *N. micropus* and *N. albigula* in having a deep,

instead of a shallow, anterointernal reentrant angle on the first upper molar and seems to belong to the *N. mexicana* group of wood rats. *Neotoma angustapalata* differs from *N. navus*, *N. mexicana*, *N. torquata*, and *N. ferruginea* in larger size, darker underparts, tail with sparse, short hairs and scaly appearance, more broadly concave sides of interpterygoid fossa at posterior end of molariform tooth-rows, larger external auditory meatus, and narrower palatine breadth. *Neotoma angustapalata* differs from the description of *N. distincta* in having a faintly bicolored tail, no ochraceous pectoral band, broadly concave sides to interpterygoid fossa, and narrower palatine breadth.

Remarks.—*Neotoma angustapalata* is represented by two specimens; the type and another specimen, an adult male, no. 37062, with skin and broken skull. The description takes into account both of these specimens. The most significant characteristics of *N. angustapalata* are its scaly-appearing tail with short, sparse hairs, dusky underparts, broadly concave sides of the interpterygoid fossa at the posterior end of molariform tooth-rows, and the narrow palatine breadth. Among named kinds of *Neotoma*, the newly named species most closely resembles *N. torquata* and *N. distincta*; however, it is geographically widely separated from these two species. *Neotoma navus* of southeastern Coahuila is the only other member of the *N. mexicana* group in northeastern Mexico.

These wood rats were taken in rocks and crevices at the base of a small hill in thick vegetation growing in deep humus. Schaldach termed the trapping site as "arid tropical tending toward humid tropical".

Measurements.—The subadult, male holotype measures as follows: Total length, 325; length of tail, 154; length of hind foot, 36; length of ear from notch, 29; basilar length (of skull), 33.9; zygomatic breadth, 22.1; interorbital breadth, 5.7; length of nasals, 15.2; length of incisive foramina, 8.7; length of palatal bridge, 8.5; least breadth of palate between first upper molars, 2.7; greatest breadth of interpterygoid space, 4.1; alveolar length of upper molariform tooth-row, 9.6. The adult, male, no. 37062, measures as follows: Total length, 380; tail vertebrae, 195; hind foot, 42; ear from notch, 31; interorbital breadth (of skull), 6.4.

Specimens examined, 2, from the type locality.

***Rattus rattus* subsp.**

Black Rat

Specimens taken, 2 from 70 km. [by highway] S Ciudad Victoria and 6 km. W of the [Pan-American] highway [at El Carrizo].

***Mus musculus* subsp.**

House Mouse

Specimen examined, 1 from 12 km. N and 4 km. W Ciudad Victoria.

***Nasua narica tamaulipensis* Goldman**

Coati

Nasua narica tamaulipensis Goldman, Proc. Biol. Soc. Washington, 55:80, June 25, 1942. (Type from Cerro de la Silla, near Monterrey, Nuevo Leon, Mexico.)

Specimen examined, 1 from 70 km. [by highway] S Ciudad Victoria and 2 km. W El Carrizo.

Remarks.—Several bands of coatis were observed in the vicinity of the village of El Carrizo. One skull of a male was obtained.

Transmitted June 8, 1951.

23-8338

Transcriber's Notes

Page [210](#): Säugthiere may be a typo for Säugethiere.

*** END OF THE PROJECT GUTENBERG EBOOK MAMMALS
FROM TAMAULIPAS, MEXICO ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, "Information about donations to the Project Gutenberg Literary Archive Foundation."
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.
- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project

Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to

you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the

efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment

including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a

copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:
www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.